

Automatic Deformation-based animation of 3D mesh

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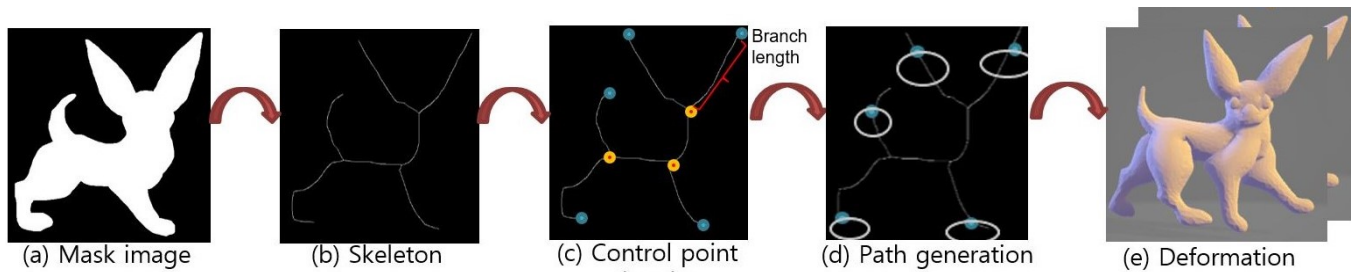


Figure 1: Animation of an 3D mesh from binary image/sketch

ABSTRACT

This work introduces a novel method for automating the object animation from a 3D mesh used in various applications such as AR/VR, gaming, etc. The 3D mesh could be a generic inflated object generated from an image or sketch. The method describes a medial axis based control point estimation to generate an animation path and perform deformation-based animations for a mesh. Our method eliminates user involvement or need for expertise in terms of rigging the mesh or region selection etc. Performance of our method shows meaningful, real-time animation of the 3D mesh. Experiments indicate that our method is more consistent and less error prone compared to existing works.

ACM Reference Format:

Lakshmi Priya Muraleedharan, Vikas Kantha Gowda, and Basavaraja Shanthappa Vandrotti. 2022. Automatic Deformation-based animation of 3D mesh. In *SA '22 Posters: SIGGRAPH Asia 2022 Posters Proceedings, December 06–09, 2022, Daegu, South Korea*. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3550082.3564184>

1 INTRODUCTION

Animating a 3D object is a time-consuming process that involves multiple steps like modelling, rigging, and animating, requiring expertise in these areas. Typically, to animate a mesh, it needs to be rigged by the designer, on top of which motion can be applied. We investigate the possibility of creating accessibility for novice users to create object animations in real-time effortlessly.

Recent works in reconstructing 3D shapes from sketches include an encoder-decoder architecture [Guillard et al. 2021] and a layering based method [Entem et al. 2019]. A real-time rigging method

was introduced for sketch-based 3D model animation by Rigmesh [Borósán et al. 2012]. A method to automatically determine 2D and 3D skinning transformations from a sparse set of controls was explored by [Jacobson et al. 2012]. In order to animate a 3D object, a user is required to select the region or point on the mesh to which the deformation is applied. Additionally, the user has to select a path for each selected point on the mesh, along which the deformation for the selected points is applied. This feedback provided by a user is not always accurate and tedious from the user's point of view. It can often lead to errors such as selecting wrong control points (Figure 3(a)) or cause overstretching resulting in unnatural animations (Figure 3(b-c)).

With this solution (Figure 1), we seek to eliminate manual intervention in the process of 3D animation which can lead to better user experience.

2 DEFORMATION BASED ANIMATION

We present an end-to-end, fully automatic method to add animation to 3D objects in real time. The pipeline follows a 3 step process as follows:

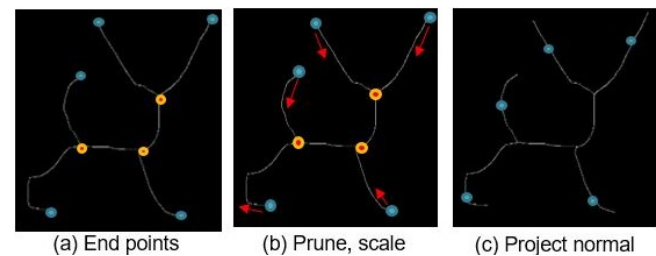


Figure 2: Various stages of control point estimation

- (1) **Control point extraction:** The pipeline enforces a medial-axis (skeleton) based approach for extracting the control points. This process takes a binary image of the 3D object as input (Figure 1(a)). The medial axis is generated by thinning this image. We then extract the end points (blue) and junction

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SA '22 Posters, December 06–09, 2022, Daegu, South Korea

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ACM ISBN 978-1-4503-9462-8/22/12.

<https://doi.org/10.1145/3550082.3564184>

points (yellow) of the medial axis (Figure 2(a)). Out of all the control points, a limited set of points are chosen based on their distance to the junction points defined as branch length (Figure 1(c)) falling within a fixed threshold. This distance is later used in generating the path for respective control points. These selected end points are then shifted inwards by a fixed distance into the skeleton (Figure 2(b)). The end points are projected onto the 3D mesh to its corresponding vertex in 3D space (Figure 2(c)).

- (2) **Path generation:** The path describes the direction in which the control point moves. This could be a discrete set of points in space for ease of calculation. Once the control points are defined, the path along which they move needs to be calculated. It is intuitive to understand that the path would be a set of points beginning with the control points and ending back at the same point as a cycle. We experimented with various simple shapes such as circles, ellipses, etc using a factor of branch length calculated earlier. It is limited by the branch length such that the distortion seems minimal and meaningful. The path can be generated in any planar orientation. However, for the ease of visualisation, we have generated results in the x-y plane (Figure 1(d)). We have also experimented with different planes based on the object category, and a simpler method to create variations in the animations is to pre-define paths based on the object category. Therefore, for each control point, an automated path is generated along which the deformation is applied.
- (3) **Deformation:** This phase follows a point-based path generation model. At each iteration, a new position for each control point is calculated based on the path and given for deformation. This stage takes in the new position of the control points in the mesh and deforms the entire mesh. We use ARAP (As Rigid As Possible) Deformation [Sorkine and Alexa 2007], which is a point-based deformation technique with no dependencies on skeletons or rigs. This technique does not require pre-defined skeletal structures or multiple view points. The ARAP method helps to deform the mesh locally based on the control point and the normal direction where the deformation needs to happen, thereby preserving the global mesh structure. When we repeat the same deformation over all the points on the path defined for a control point, the mesh gets animated in a cycle. This deformation happens simultaneously for all the selected control points based on their respective paths. For each animation frame, the deformed mesh is written to a GLTF file.

3 RESULTS & DISCUSSION

Our method produces smooth and rigid 3D animations with consistent results. ARAP deforms the mesh locally with minimum distortion globally. The user is not involved in the process of control point selection or path generation. This makes the entire process of animating the 3D mesh created from inflating the image or sketch completely automatic.

Figure 1 shows the overall end-to-end pipeline of deformation-based animation. The input to the pipeline is an image or sketch whose binary mask is estimated, and the output is the deformation

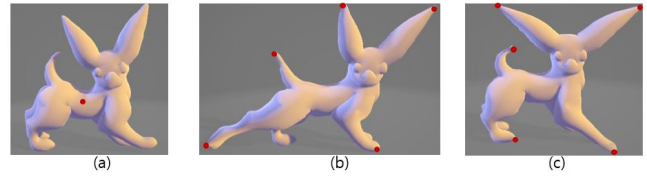


Figure 3: Unnatural deformation due to user overstretching the mesh

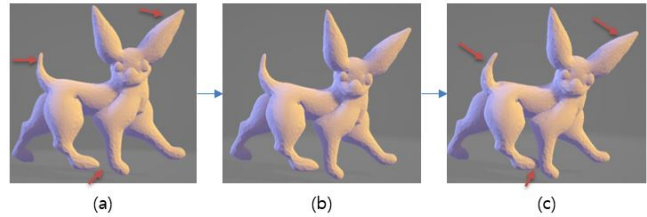


Figure 4: Result depicting different frames after mesh deformation (our method)

of the inflated meshes. The multiple frames of the deformed mesh are shown in the final stage of the pipeline, which are stored in GLTF format (Figure 4). We have tested our method on random sketch-like images from the internet as well as publically available 3D mesh datasets. Our experiments resulted in meaningful and consistent performance by our method. Our entire pipeline takes only 18 s for generating animations, whereas original ARAP based deformation takes 50 s on average for a new user and 35 s for an expert user to define the animation alone, for the same input image and deformation given in 1(a).

4 CONCLUSION

In this work, we realised a method for automating the animation of 3D mesh models by reducing the overhead of rigging the mesh model or selecting the control points that need to be moved and providing a path for animation. As a result, the user is not involved in any stages of animating the model. In future, we will extend this method to make use of it in video retargeting of mesh models.

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